

Exploratory Data Analysis of Online Retail Dataset

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Tools Used: Python (Pandas, Numpy, Matplotlib, Seaborn)

Dataset: OnlineRetail Dataset

Business Insights

1. Sales declined in January, April, and July, while peak sales were recorded in February, May, and September.
2. The top three purchasing countries (UK excluded) by sales were Germany (92,911), France (84,222), and EIRE (75,349). The least-performing countries included Bahrain (218), Czech Republic (160), and Saudi Arabia (120).
3. The product *"Jumbo Bag Red Retrospot"* (9,854) recorded the highest quantity ordered, while *"Hanging Ridge Glass T-Light Holder"* (-24) recorded the lowest quantity sold due to product returns, resulting in negative values.
4. Weak negative correlation (-0.26) exists between unit price and quantity ordered, indicating that lower prices tend to encourage higher purchase volumes.
5. Sales and quantity ordered showed a strong positive correlation (0.69), suggesting that higher order quantities significantly contribute to increased sales revenue.

Recommendations

1. Increase inventory levels and staffing during peak sales periods. For example, promote gift items in February for Valentine's Day, children-related products in May, and seasonal decorations toward September in preparation for Halloween.
2. Implement targeted marketing campaigns in low-performing regions such as Bahrain, the Czech Republic, and Saudi Arabia to improve customer engagement and sales.
3. Maintain higher inventory levels for high-demand products such as *"Jumbo Bag Red Retrospot."* Additionally investigate the causes of low demand and product returns associated with *"Hanging Heart T-Light Holder."*
4. Introduce promotional pricing and discount strategies where appropriate, as lower unit prices appear to encourage increased purchasing volume.

EDA Keynotes

The Online Retail dataset underwent extensive cleaning, during which approximately 30% of the rows were removed due to missing values, duplicates, outliers, and invalid zero-value entries.

The United Kingdom was excluded from the country-level sales comparison because its exceptionally high sales value (3,021,709.224) would create significant bias and reduce the comparability of other countries.

The findings suggest that the company can improve overall performance through:

- Increased inventory during peak sales periods
- Stronger regional marketing strategies
- Greater availability of high-demand products
- Improvements to low-performing products
- Strategic promotional discount campaigns